

# Combinations of measurements that are simultaneous fits of parameters of interest and systematic uncertainties using the BLUE method

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ABSTRACT: Abstract...

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## 1 Introduction

Measurements of physics parameters, such as the Higgs boson mass, Higgs boson coupling, top quark mass, the strong coupling constant and the parameters of the neutrino mixing matrix are at the heart of experimental particle physics. The existence of multiple experiments, for example, at the Large Hadron Collider (LHC) [1] and in the neutrino sector, motivates the need to combine measurements from different experiments and from different measurements within single experiments. One technique for performing such combinations is the Best Linear Unbiased Estimate (BLUE) [2, 3], which requires the covariance of the measurements being combined. This technique has been widely used on analyses where the physics parameters of interest (POIs) were extracted from a likelihood fit where only the POIs are free parameters (referred to as a statistical-only likelihood fit in the remainder of the paper), see e.g. Refs [4–7]. Systematic uncertainties are typically evaluated by repeating the fits with varied models that represent the systematic uncertainties [8]. The systematic uncertainties can then be added in quadrature with the statistical uncertainty from the fit to give the total uncertainty. Systematic uncertainties are normally assumed to be uncorrelated with each other and the covariance between any two measurements is then straightforward to estimate once the correlation between systematics in different measurements is established.

A feature of the LHC analysis era is that many measurements of physics parameters are extracted from likelihood fits that include both the physics parameters of interest and a set of parameters which represent the systematic uncertainties, often referred to as nuisance parameters (NP). The results of such fits (referred to as full-likelihood fits in the following) are estimates of both the POIs and the NPs and a corresponding covariance matrix. As

well as conveniently including the systematic uncertainties into the likelihood, this means the data can add to the knowledge on the NP and reduce the uncertainty in the POIs. Recent examples include measurements of the  $t\bar{t}$  cross-section near threshold [9, 10]. If the full-likelihood is available, such measurements can conveniently be combined by forming the joint likelihood of the measurements (see e.g. Ref. [11]); however, in many cases the full-likelihoods are not publicly available and hence combination techniques starting from the results of the fits are needed. The NPs after a full-likelihood fit will typically have non-zero correlation, which at first glance appears to break the view of systematic uncertainties as being uncorrelated with each other. Ref. [12] proposes an approximate method and tool (Convino) for combining such measurements, indicating that the BLUE method cannot be used without information beyond the covariance matrix and central values of the full-likelihood fits. This method has subsequently been used in several recent publications [13–16]. While the latter three combine statistical-only likelihood fits (in which case the Convino method corresponds to BLUE), Ref. [13] combines a CMS full-likelihood fit with an ATLAS statistical-only likelihood fit. Ref. [17] details how the contributions of the systematic uncertainties to measurements from full-likelihood fits can be obtained from the covariance matrix if the measurement is in the Gaussian regime<sup>1</sup>. The BLUE method can then be used to combine such measurements based on the normal assumption that the systematic uncertainties are uncorrelated with each other. The authors of Ref. [17] noted that it is possible to combine either the POIs or the POIs and the NPs from full-likelihood measurements, but showed for the example given in the paper that the two options gave the same precision. In this paper, we show that the combination of POIs and NPs can, in certain situations, result in a smaller uncertainty on the combined POIs than when only the POIs are combined. We demonstrate that this behaviour occurs even if only one of the input measurements is a full-likelihood measurement. Two examples are investigated, one purely hypothetical and the second combining two recent ATLAS top quark mass measurements. For the first example, we also perform a combination with the Convino tool [12] and show with pseudo-data that the uncertainty from this method is underestimated. The paper is accompanied by a software tool implementing the logic described in the paper [1].

The paper is structured as follows: in Section 2 we review the mathematical framework for combining multiple full-likelihood analyses. In Section 3, we examine how statistical-only likelihood fits can be incorporated into the setup and show that the NP measurements from a full-likelihood analysis can improve the combined estimate of POIs even when one full-likelihood analysis is combined with one statistical-only likelihood analysis. In Section 4, we set up two hypothetical analyses using full-likelihood fits. The analyses are combined with BLUE using the POIs only and using the POIs and NPs. The results are compared with a combination using Convino and are validated using pseudo-experiments. In Section 5, we combine two recent ATLAS top quark mass measurements and show that including the NPs reduces the uncertainty on the combined top quark mass estimate. We finally summarise the paper in Section 6.

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<sup>1</sup>For measurements not in the Gaussian regime, the contributions of the systematic uncertainties can be obtained from varying the global observables (see Section 4 for the definition of the global observables).

## 2 Mathematical framework

The output of a full-likelihood analysis that is in the Gaussian regime is  $M$  estimates of the POIs and  $N$  estimates of the NPs, which we collectively denote as  $\vec{\theta}$  where the best fit values for the parameters  $\vec{\theta}$  are complemented with the covariance matrix  $C$  that encodes the uncertainties on the parameters and the correlations between them. To combine this fit with another fit where at least some of the POIs and NPs are in common, we need to estimate the correlation between the estimates from each analysis. For many cases, the two analyses will have been performed on orthogonal datasets (e.g. analyses from different experiments or analyses within the same experiment that use selections that ensure orthogonality). In these cases, the correlation between the fitted parameters arises from the common sources of systematic uncertainties. Following [17], the contribution to the uncertainty on each parameter from every source of systematic uncertainty can often be estimated from the covariance matrix  $C$  and we denote the contribution of uncertainty source  $k$  on the estimate of parameter  $i$  as  $\Gamma_{ik}$ . The covariance between two analyses  $A$  and  $B$  can then be written as:

$$C_{AB} = \begin{pmatrix} C_A & D \\ D & C_B \end{pmatrix} \quad (2.1)$$

where the elements of the cross-correlation matrix are  $D_{ij} = \Gamma_{ik}\Gamma_{jk}$ <sup>2</sup> and  $C_A$  and  $C_B$  are the covariance matrices for the two analyses  $A$  and  $B$ . In the case that the analyses  $A$  and  $B$  have some statistical correlation (for example they use overlapping sets of events), the statistical covariance can be estimated using pseudo-experiments on the individual analyses to produce a statistical covariance matrix  $S$  and the cross-correlation matrix must then include those cross-correlations with  $D_{ij} = \Gamma_{ik}\Gamma_{jk} + S_{ij}$  where  $S_{ij}$  is the element of  $S$  that expresses the statistical covariance between the estimates  $i$  and  $j$ . As we are dealing with Gaussian measurements, the best-fit values for the parameters are obtained by minimising the  $\chi^2$

$$\chi^2 = (\vec{\hat{\theta}} - \vec{\theta})^T C_{AB}^{-1} (\vec{\hat{\theta}} - \vec{\theta}) \quad (2.2)$$

where the vectors  $\vec{\hat{\theta}}$  contain the best fit values for all the parameters in analyses  $A$  and  $B$ . The minimisation of this  $\chi^2$  can be done analytically and corresponds to the well-known BLUE method [2, 3]. The combined estimate for a parameter  $\theta_\alpha$  is

$$\theta_\alpha = (U^T C_{AB}^{-1} U)_{\alpha j}^{-1} (U^T C_{AB}^{-1})_{ji} \hat{\theta}_i \quad (2.3)$$

where the  $U$  matrix is defined such that  $U_{i\alpha} = 1$  if  $i$  is an estimate of parameter  $\alpha$  and zero otherwise. As discussed in Ref. [3], the combined estimate of each parameter receives contributions from all parameters which are estimated by both  $A$  and  $B$ . The weight of an

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<sup>2</sup>In the paper, unless otherwise stated, we use the Einstein summation convention where repeated indices denote a sum over that index.

estimate  $\hat{\theta}_i$  on the combined estimate  $\theta_\alpha$  is  $\lambda_{\alpha i} = (U^T C_{AB}^{-1} U)_{\alpha j}^{-1} (U^T C_{AB}^{-1})_{ji}$  and is referred to as the BLUE weight. These weights obey the constraints:

$$\sum_i^{\text{estimates of } \alpha} \lambda_{\alpha i} = 1 \quad (2.4)$$

$$\sum_i^{\text{estimates not of } \alpha} \lambda_{\alpha i} = 0. \quad (2.5)$$

The estimates of the NPs from the two analyses  $A$  and  $B$  will therefore contribute to the combined estimates of the POIs and the covariance matrix of the combined estimates is:

$$c_{\alpha\beta} = (U^T C_{AB}^{-1} U)_{\alpha\beta}^{-1}. \quad (2.6)$$

It is a choice to include the estimates of the NPs in the above procedure; it is also possible to choose to only consider the POIs from the two analyses. The mathematics remains the same (as it makes no distinction between POIs and NPs) and the covariance matrix  $C_{AB}$  reduces in size to be only  $2M \times 2M$ , rather than  $(2M + 2N) \times (2M + 2N)$ . The resulting combined estimates will only have contributions from the POIs. As this reduced combination uses less data, it follows from standard information theory [18] that the uncertainty on the POIs obtained from combining both the POIs and the NPs is less than or equal to the uncertainty on the POIs obtained from combining only the POIs. We examine specific examples of this situation in Section 4.

### 3 Combining non-profiled analyses

The framework described above imagines analyses that have performed a full-likelihood fit to obtain estimates of the POIs and the NPs. There are many measurements where the fit to the data uses only the statistical uncertainties and the systematic uncertainties are propagated through the measurement. The outputs of such a measurement are the best-fit value of the POIs, the covariance between them ( $O$ ) and the contribution of each systematic uncertainty  $k$  to each POI  $i$ ,  $\Gamma_{ik}$ . The prior measurements of the NPs can be thought of as having a best-fit value of zero and uncertainty one that is uncorrelated with all other NPs. This allows us to enlarge the covariance matrix of a non-profiled measurement to:

$$C = \begin{pmatrix} O & \Gamma \\ \Gamma^T & I \end{pmatrix} \quad (3.1)$$

where  $I$  is the identity matrix. With this matrix in hand, a non-profiled analysis can be included in exactly the same way as a profiled analysis into the combined estimates discussed in Section 2.

We now apply this setup to profiled-likelihood analysis  $A$  and a non-profiled analysis  $B$  which each have one POI  $p$  and  $N$  NPs  $\vec{n}$ . Denoting the two estimates of the POI as  $p_A$  and  $p_B$ , the estimates of the NP from analysis  $A$  as  $\vec{n}_A$  and the prior estimates of the

NP implicit in analysis  $B$  as  $\vec{n}_B = \vec{0}$  the combined estimate of the POI  $p$  obtained with Equation 2.3 is:

$$\begin{aligned} p &= \lambda_{p_A} p_A + \lambda_{p_B} p_B + \sum_{i=0}^N (\lambda_{n_{A,i}} n_{A,i} + \lambda_{n_{B,i}} n_{B,i}) \\ &= \lambda_{p_A} p_A + (1 - \lambda_{p_A}) p_B + \sum_{i=0}^N \lambda_{n_{A,i}} n_{A,i} \end{aligned} \quad (3.2)$$

where  $\lambda_{p_j}$  are the BLUE weights for the POIs from measurement  $j$  and  $\lambda_{n_{j,i}}$  are the BLUE weights for NP  $i$  from measurement  $j$ . We observe that the combined estimate is now comprised of the estimates of the two POIs plus a contribution from the estimates of the NPs from measurement  $A$ . One way to view this is that the estimates of the NPs from analysis  $A$  are being used to improve the estimate of the POI from analysis  $B$ . In this way, the combined estimate of the POI has a smaller variance than if one were to combine only the two POIs from the two analyses. This behaviour is demonstrated in the combinations described in Sections 4 and 5.

#### 4 Example combination 1

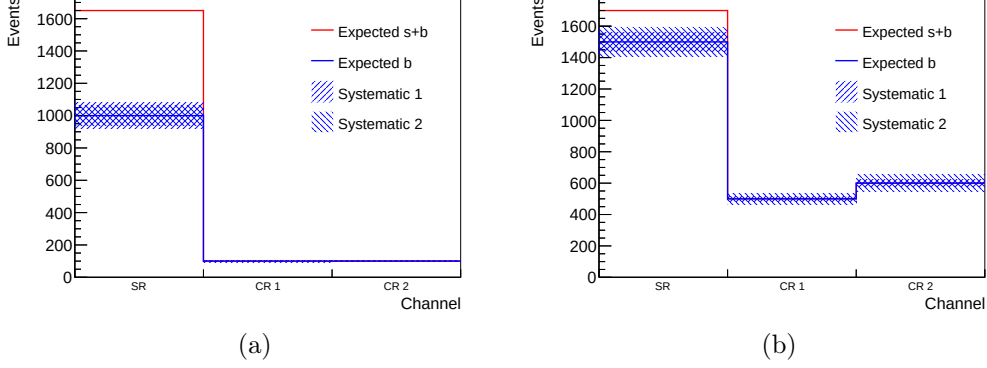
To illustrate the features of the combined estimates discussed in Section 2, we consider two example analyses that are to be combined. The two analyses both seek to measure the rate of a signal process,  $\mu$ , where  $\mu = 1$  is defined to be the nominal prediction for the signal. There is a background process which is subject to systematic uncertainties. Both analyses use a single signal region (SR) which contains both signal and background and two control regions (CR) which contain only background events. A full-likelihood fit is performed in each analysis to the signal and control regions, where the events in the control regions provide information on the NPs. The likelihood in each analysis is constructed as:

$$\mathcal{L}(d|\mu, \theta_1, \theta_2) = P(d_{SR}|s(\mu) + b_{SR}(\theta_1, \theta_2)) \prod_i^{CR} (d_{CRi}|b_{CRi}(\theta_1, \theta_2)) \prod_s G(g_i|\theta_i, \sigma = 1) \quad (4.1)$$

where  $d_{SR}$  and  $d_{CRi}$  are the number of data events in the signal region and two control regions  $i = 1, 2$ ,  $P$  denotes a Poisson probability distribution function (PDF) and  $G$  a Gaussian PDF. The number of expected signal events in the SR is  $s(\mu) = s_0\mu$ , where  $s_0$  is the nominal expectation for the signal. The number of expected background events in the SR and CRs are  $b_{SR}$  and  $b_{CRi}$ . The final term contains the Gaussian constraints on the systematic uncertainties, where  $g_i$  are referred to as the global observables and have a nominal value of zero. The Asimov dataset [19] is defined to be the expected number of events for  $\mu = 1$  and  $g_i = 0$ .

Analysis  $A$  is set up such that it has high sensitivity to  $\mu$  but low constraining power for the NP. Analysis  $B$  is set up such that it has low sensitivity to  $\mu$  but high constraining power on the NPs. The expected signal and background events for the two analyses are shown in Figure 1. In order to test a combination of a non-profiled measurement, the fit for analysis  $A$  is also performed with only the SR. In this configuration, there is one data bin

and one free parameter ( $\mu$ ) and so the NPs cannot be constrained. The covariance matrix from this fit has exactly the form shown in Equation 3.1, allowing for a test where analysis  $A$  is effectively a statistical-only likelihood.



**Figure 1:** The expected number of background events (blue) and signal plus background (red) is shown for (a) analysis  $A$  and (b) analysis  $B$ . The sizes of the hashed areas show the uncertainties in the expected number of background events that originate from the systematic uncertainties. The signal distribution assumes  $\mu = 1$ .

The results of the profile-likelihood fits to the Asimov dataset for the two analyses are shown in Table 1. Analysis  $A$  has significantly better precision on the POI than analysis  $B$ , while analysis  $B$  has a more precise estimate of systematic 2. The combination using only the POIs makes only a mild improvement on the result from analysis  $A$ , due to the relatively poor precision of analysis  $B$ . Including the estimates of the NPs into the combination significantly improves the precision of the POI. This is an example of where performing the combination across the full set of estimated parameters improves on the combination of only the POIs. The origin of this improvement can be understood by examining the BLUE weights of the combinations, which are shown in Table 2. In the combination of the POIs and NPs, the estimates of the second NP carry substantial weight in the combined estimate of the POI and the precision of this estimate is improved by including the estimates of the NPs. Table 1 also shows the expected uncertainty for a combination using results from a fit on only the SR from analysis  $A$  and the SR+CR from analysis  $B$ . This corresponds to combining a statistical-only fit with a full-likelihood fit and allows us to test the setup discussed in Section 3.

The validity of the combinations is tested by performing pseudo-experiments on the analyses. The global observables  $g_i$  are fluctuated according to Gaussian distributions with mean zero and width one. For each pseudo-experiment, the values of  $g_i$  are common to both analyses (representing correlated systematic uncertainties). In each analysis, the number of events in each signal and control region is fluctuated according to a Poisson distribution. In each pseudo-experiment, the profile-likelihood fit for each analysis is performed and the outputs of the fits are then used to perform the same two combinations performed above on the Asimov datasets. The means of the combined POIs and NPs are found to

| Measurement                                               | POI  | Syst1 | Syst2 |
|-----------------------------------------------------------|------|-------|-------|
| Analysis 1 SR only                                        | 0.13 | 1.00  | 1.00  |
| Analysis 1                                                | 0.13 | 0.83  | 0.91  |
| Analysis 2                                                | 0.28 | 0.92  | 0.50  |
| Combination of analysis 1 & 2, using POI & NP             | 0.10 | 0.82  | 0.47  |
| Combination of analysis 1 & 2, using only POI             | 0.12 | -     | -     |
| Combination of analysis 1 SR & analysis 2, using POI & NP | 0.11 | 0.89  | 0.49  |

**Table 1:** Uncertainties obtained from the profile-likelihood fits to the two example analyses and the uncertainties obtained by combining the two analyses. The first combination uses the estimates of both the POIs and NPs from both analyses, while the second combination uses only the estimates of the POI from the two analyses.

|                           | POI 1 | POI 2 | NP 1,1 | NP 1,2 | NP 2,1 | NP 2,2 |
|---------------------------|-------|-------|--------|--------|--------|--------|
| Combination of POI and NP | 1.02  | -0.02 | 0.03   | -0.03  | 0.10   | -0.10  |
| Combination of POI only   | 0.89  | 0.11  | -      | -      | -      | -      |

**Table 2:** BLUE weights for the two combinations of the example analyses.

| Combination method   | Reported uncertainty of combination | RMS of combined $\mu$ |
|----------------------|-------------------------------------|-----------------------|
| BLUE with POI only   | 0.12                                | 0.12                  |
| BLUE with POI and NP | 0.10                                | 0.10                  |
| Convino              | 0.10                                | 0.12                  |

**Table 3:** Comparison of the uncertainty from different combination methods and the RMS of the combined  $\mu$  derived from pseudo-experiments.

be consistent with one and the RMS agrees with the expected precision shown in Table 1 for all combinations. This validates that the combination procedure is unbiased and has a reliable uncertainty estimate.

#### 4.1 Comparison with Convino

The Convino method presented in [12] presents an alternative method for combining profile-likelihood analyses. It is noticeable that while that method uses the full covariance matrix from each input analysis and the estimates of the POIs, the best-fit estimates for the NPs from each analysis are not used. The combination of analysis A and B is performed with the Convino method and the method presented here. Pseudo-experiments are performed as described in the previous section and the RMS of the combined  $\mu$  from the pseudo-experiments is shown in Table 3. While the uncertainties from the BLUE method match the RMS from the pseudo-experiments, the uncertainty from Convino is found to be underestimated compared to the RMS from the pseudo-experiments. It is expected that this is because Convino is not using the best-fit estimates for the NPs in the approximate likelihood. This behaviour is discussed in more detail in Appendix A.

## 5 Example combination 2

The example combination in the previous example serves to illustrate the features of combining both the POIs and NPs from full-likelihood analyses, but it uses analyses that are significantly simpler than realistic measurements. To test the method in a more realistic environment, we use two recently published ATLAS measurements of the top quark mass ( $m_t$ ). The first measurement uses  $t\bar{t}$  events with a high transverse momentum (boosted) top quark [20]. This analysis is a full-likelihood fit with non-negligible constraints on the NPs. The likelihood and post-fit covariance matrix are publicly available in HepData [21] and the latter provides the first input for the second example combination. The second input for this example combination is the ATLAS measurement of the top quark mass using  $t\bar{t}$  events where a  $J/\psi$  is produced from one of the top quark decays [22]. This analysis uses a statistical-only likelihood but shares several of the uncertainties in the boosted measurement and so the combination can explore the properties discussed in Section 3. The full breakdown of uncertainties is stored in HepData [23] and this is used to perform the combination. The  $J/\psi$  analysis has a large systematic uncertainty originating from the choice of recoil model in the  $t\bar{t}$  Monte Carlo simulation. The NP for this uncertainty is constrained in the boosted analysis and hence we expect that including the measurements of this NP into the combination will improve the overall precision.

To perform this combination, it is necessary to decide if each of the systematic uncertainties in the two measurements has an equivalent uncertainty in the other measurement and hence is correlated. A fully accurate treatment of the correlations requires dedicated efforts by the ATLAS collaboration (similar to those done for e.g. Ref [4]). In this paper, we aim only to have a reasonable enough correlation model so that we can explore the statistical features of the combination discussed in Section 3. As the two measurements use the same ATLAS dataset, the detector uncertainties largely have a clear mapping - uncertainties that have the same name are assumed to be equivalent. One exception is the jet energy scale (JES) flavour uncertainties, where the boosted analysis uses a more detailed procedure. The  $b$ -JES uncertainties in the  $J/\psi$  analysis are assumed to be uncorrelated with the three heavy-flavour JES uncertainties in the boosted analysis. This choice has negligible impact on the combination as the JES uncertainties are very small in the  $J/\psi$  analysis. Similarly, many of the  $t\bar{t}$  and single top modelling uncertainties are derived from the same procedures and in these cases we also assume the uncertainties to be correlated between the two measurements. This includes the systematic uncertainty for the recoil model, which is the largest single uncertainty in the  $J/\psi$  analysis. The parton-shower and hadronisation uncertainty is decorrelated between the three observables used in the fit for the boosted measurement. We chose to correlate the uncertainty in the  $m_t$  sensitive variable with the corresponding uncertainty in the  $J/\psi$  analysis. Other choices make a negligible difference to the combination **NEED TO ACTUALLY CHECK THIS**. Uncertainties in the background estimates for the two analyses are considered to be uncorrelated due to the very different phase-spaces targeted by the measurements. Any systematic uncertainties that appear in only one of the measurements (e.g. the soft-muon systematic uncertainties in the  $J/\psi$  measurement) are assumed to be uncorrelated with all uncertainties in the sec-

| Combination       | BLUE weights |          |           |           | $\Delta m_t$ (GeV) |
|-------------------|--------------|----------|-----------|-----------|--------------------|
|                   | $m_t$        | $m_t$    | Recoil NP | Recoil NP |                    |
|                   | boosted      | $J/\psi$ | boosted   | $J/\psi$  |                    |
| POI only          | 0.90         | 0.10     | -         | -         | 0.50               |
| POI and recoil NP | 0.84         | 0.16     | -0.17     | 0.17      | 0.48               |

**Table 4:** BLUE weights and combined uncertainty in  $m_t$  ( $\Delta m_t$ ) for the  $m_t$  combinations of the ATLAS boosted and  $J/\psi$  measurements.

ond measurement. The overall correlation between the  $m_t$  estimators for the boosted and  $J/\psi$  analyses is found to be 0.01.

The combination is first performed using only the two POI measurements (of  $m_t$ ). This yields a combined measurement of  $m_t$  with an uncertainty of 0.50 GeV. The BLUE weights are shown in Table 4 and the boosted measurement is seen to dominate the combination. The combination is then performed using the two POI measurements and the two measurements of the recoil NP. This yields a combined measurement of  $m_t$  with an uncertainty of 0.48 GeV, improving on the combination using only the POIs. The corresponding BLUE weights are shown in Table 4. The measurement of the recoil NP from the boosted measurement enters with significant weight, which can be viewed as propagating the information on this NP to the  $J/\psi$  measurement as anticipated in Section 3. The weight of the  $J/\psi$  analysis is seen to increase when the recoil NP is added to the combination, which is consistent with the  $J/\psi$  analysis effectively having higher precision once the information on the recoil NP is available.

This combination of the two ATLAS  $m_t$  measurements provides a realistic example of where including the measurements of the NP into a combination can increase the precision on the POI. The decision of which NP to include in a real combination of a parameter like  $m_t$  should consider both the potential increase in the precision and the extent to which it is reasonable to propagate the information on the NPs across different measurements.

## 6 Summary

In this paper, we consider combinations of measurements that have been performed with full-likelihood or statistical-only likelihood fits. The work of Ref. [17] sets out how to obtain the breakdown of uncertainties for full-likelihood fits and makes their inclusion in BLUE combinations straightforward. We have shown through two examples that BLUE combinations that use both the parameters of interest (POIs) and the nuisance parameters can improve on combinations that use only the POIs. In the first example, we also verify with pseudo-experiments that the combined estimates are unbiased with reliable uncertainty estimates. The degree of improvement will depend largely on the input measurements and this can easily be explored through the software tool that accompanies the paper. We have also shown that the Convino tool proposed in Ref. [12] can underestimate the uncertainty on a physics parameter when it is applied to analyses using full-likelihood fits. For cases where it is not possible to make a full-likelihood combination, the BLUE method outlined

here provides a reliable way for combining measurements of physics quantities irrespective of how the original measurements are performed.

## A The minimum of the Convino $\chi^2$

The first assumption in Ref. [12] for the method used in the Convino tool is that the likelihood of a measurement can be approximated as<sup>3</sup>:

$$\chi^2 = (d_\mu - x_\mu - k_{\mu i} \lambda_i) M_{\mu\nu} (d_\nu - x_\nu - k_{\nu j} \lambda_j) + \lambda_i D_{ij} \lambda_j + \lambda_i \lambda_i, \quad (\text{A.1})$$

where  $x_\mu$  are the POIs,  $d_\mu$  are the estimates of those POIs from this measurement and  $\lambda_i$  are the NPs. It is worth noting that the estimates of the NP from the measurement do not appear in the approximate likelihood. The matrices  $M$ ,  $D$  and  $k$  are defined in Ref. [12], but the precise definition is not important in the following, we only need to know that  $M$  and  $D$  are symmetric matrices. The minimum of this  $\chi^2$  is found by solving for  $\frac{\partial \chi^2}{\partial x_\gamma} = 0$  and  $\frac{\partial \chi^2}{\partial \lambda_\epsilon} = 0$ . The first requirement yields:

$$\frac{\partial \chi^2}{\partial x_\gamma} = -2(d_\mu - x_\mu - k_{\mu i} \lambda_i) M_{\mu\gamma} = 0 \quad (\text{A.2})$$

which for non-zero  $M$  is solved with

$$(d_\mu - x_\mu - k_{\mu i} \lambda_i) = 0. \quad (\text{A.3})$$

The second requirement yields:

$$\frac{\partial \chi^2}{\partial \lambda_\epsilon} = -2(d_\mu - x_\mu - k_{\mu i} \lambda_i) M_{\mu\epsilon} + 2D_{\epsilon j} \lambda_j + 2\lambda_\epsilon = 0. \quad (\text{A.4})$$

Substituting Equation A.3 into Equation A.4 yields  $2D_{\epsilon j} \lambda_j + 2\lambda_\epsilon = 0$ , which is solved with  $\lambda_\epsilon = 0$ . The minimum of the  $\chi^2$  is therefore:

$$\lambda_\epsilon = 0 \quad (\text{A.5})$$

$$x_\mu = d_\mu. \quad (\text{A.6})$$

The values of the POIs at the minimum  $\chi^2$  correctly correspond to the measured values but the values of the NPs at the minimum  $\chi^2$  are always 0. For a full-likelihood measurement the actual best-fit values of the NPs are not at zero and hence this approximate likelihood is not appropriate in many cases.

The only case where it appears that the approximation is valid is where the measurement has no constraining power on the NPs, in which case the NPs must fit at 0. As discussed in Ref. [12] the matrix  $D$  is then zero and the approximate likelihood simplifies to:

$$\chi^2 = (d_\mu - x_\mu - k_{\mu i} \lambda_i) M_{\mu\nu} (d_\nu - x_\nu - k_{\nu j} \lambda_j) + \lambda_i \lambda_i. \quad (\text{A.7})$$

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<sup>3</sup>For brevity, we restrict ourselves here to the Neyman  $\chi^2$  and absolute systematic uncertainties.

This is form of  $\chi^2$  where  $M$  is the inverse of the statistical covariance matrix and  $k_{\nu j}$  is the impact of systematic source  $j$  on parameter estimate  $\nu$  and is known [3] to be entirely equivalent to:

$$\chi^2 = (d_\mu - x_\mu)C_{\mu\nu}(d_\nu - x_\nu) \quad (\text{A.8})$$

where  $C$  is the inverse of the full covariance matrix and the minimum of this  $\chi^2$  corresponds to the BLUE combination of the POIs. The equivalence between Convino and BLUE for analyses without constraints on the NP has been seen numerically in previous results [4, 14–16]. The results in Section 4.1 show that the Convino tool can underestimate the combined uncertainties on the POIs when combining measurements with constraints on the NPs. We conclude that the approximation in the Convino tool is only appropriate for measurements with no constraints on the NPs and it should not be used to combine full-likelihood analyses.

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